

— PROGRESS MEASUREMENT & EVM • MODERN PROJECT CONTROLS SERIES

1ST
EDITION
2026

PROJECT PROGRESS MEASUREMENT & EARNED VALUE MANAGEMENT

A PRACTICAL GUIDE FOR PROJECT CONTROLS PROFESSIONALS



MEASURE
PERFORMANCE



MANAGE
VARIANCE



FORECAST
OUTCOMES



DRIVE
SUCCESS



ORACLE
PRIMAVERA P6

 Power BI

 Microsoft
Project

 EVM
BEST PRACTICES



APEX BUSINESS SOLUTIONS
EMPOWERING PROJECT CONTROLS EXCELLENCE



SCAN TO LEARN MORE ►

PRACTICAL.
ACTIONABLE.
RESULTS.

Project Progress Measurement & Earned Value Management

A Practical Guide for Project Controls Professionals

Copyright

Copyright © 2026 Apex Business Solutions. All rights reserved.

First Edition, 2026

ISBN: 978-X-XXXX-XXXX-X (Print) | ISBN: 978-X-XXXX-XXXX-X (eBook)

Published by Apex Business Solutions

Legal Notice

No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means—electronic, mechanical, photocopying, recording, or otherwise—without prior written permission from the publisher, except for brief quotations in critical reviews or articles.

Disclaimer

While every effort has been made to ensure accuracy, the information in this book is provided "as is" without warranty of any kind. The author and publisher disclaim any liability for errors, omissions, or damages arising from the use of this information.

Trademark Notices

Oracle, Primavera, Microsoft Project, Power BI, and other product names are trademarks of their respective owners. This publication is independent and not affiliated with or endorsed by these companies.

Contact: apexbusiness.training@gmail.com

Website: <https://www.facebook.com/primaveraasia/>

APEX BUSINESS SOLUTIONS

MODERN PROJECT CONTROLS SERIES · VOLUME 3

Project Progress Measurement & Earned Value Management

A Practical Guide for Project Controls Professionals

For Primavera P6 · Oracle Primavera Cloud · Microsoft Project · Asta Powerproject

and any scheduling platform

First Edition · 2026

apexbusiness.training@gmail.com

Disclaimer

This book provides general guidance on project progress measurement and earned value management practices. It is intended for educational and informational purposes only and does not constitute professional consulting advice.

No Professional Relationship

Reading this book does not create a consultant-client relationship. Readers should consult qualified project controls professionals for advice specific to their projects.

Industry Standards

While this guide references industry best practices, standards evolve. Readers should verify current standards and contractual requirements for their specific projects.

Software and Tools

Software features and capabilities described are current as of 2026. Users should consult official documentation for their specific software versions.

Limitation of Liability

The author and publisher shall not be liable for any loss of profit or commercial damages, including but not limited to special, incidental, consequential, or other damages, arising from the use of information in this book.

About This Guide

This guide is written for project controls professionals — planning engineers, cost engineers, project engineers and project managers — who need progress numbers that survive scrutiny: from site data collection through weighting systems, earned value analysis and S-curves to modern dashboards and AI applications. The methods are platform-agnostic; where software steps are shown, Primavera P6 is used as the reference tool.

How to read it: Chapters 1–4 build the foundation (the control cycle, measurement methods, weighting systems). Chapters 5–8 are the machinery (P6 implementation, EVM with worked examples, S-curves, progress rules). Chapters 9–12 apply it phase by phase, ending in the ready-to-use measurement matrix. Chapters 13–17 modernise the practice: dashboards, AI, owner-grade governance, case studies and the template library.

Look for the recurring boxes: Apex Tips (practical advice), Common Mistakes (what to avoid), Industry Best Practice (the professional standard), and AI Insight (where technology genuinely helps).

Copyright © 2026 Apex Business Solutions. All rights reserved.

No part of this publication may be reproduced, stored or transmitted in any form without written permission from the publisher. Oracle and Primavera are registered trademarks of Oracle Corporation. Microsoft Project and Power BI are trademarks of Microsoft Corporation. Asta Powerproject is a trademark of Elecosoft. This publication is not affiliated with or endorsed by any of these companies.

Contents

Chapter 1: Introduction to Progress Measurement

Every month, on thousands of projects around the world, someone stands in front of management and says "we are 62% complete." On well-run projects, that number is the output of a disciplined measurement system — traceable, verifiable, and consistent from month to month. On most projects, it is a negotiation between what the site believes, what the contractor wants to invoice, and what the client is willing to accept. The difference between those two worlds is the subject of this book.

Progress measurement is the process of determining, objectively and repeatably, how much of a project's scope has been completed at a point in time. It sounds administrative. It is not. Progress measurement is the foundation on which every other project control function stands: without a trustworthy progress figure, your schedule forecast is fiction, your earned value metrics are noise, your invoicing is a dispute waiting to happen, and your management is making decisions in the dark.

Why progress measurement fails

After decades of industry experience across oil & gas, EPC and infrastructure projects, the same failure patterns appear again and again:

- 1. Progress by feel.** The most common failure. A supervisor estimates "about 80%" based on visual impression. Human beings systematically overestimate completion of work in progress — the industry joke that a project spends six months at 90% exists because the last 10% of work was never actually 10%.
- 2. No agreed measurement basis.** The contractor measures one way, the client another, and the difference surfaces as a payment dispute in month five — far more expensive to resolve than it would have been to prevent in week one.
- 3. Front-loaded weightages.** Deliberately or carelessly, early work carries inflated weight. Progress and payment race ahead of physical reality, then flatline while the hard, under-weighted work grinds on.
- 4. Progress without evidence.** Percentages claimed with no backup — no quantity records, no signed inspection reports, no document register. Unverifiable progress is unclaimable progress when challenged.
- 5. Measuring effort instead of output.** Hours spent is not work completed. A crew can burn 10,000 hours producing very little; expended manhours tell you about cost, not progress.
- 6. Systems that don't survive contact with the site.** Elaborate measurement procedures that require data the site cannot realistically provide every week. The system silently degrades into estimation within two months.

⚠️ COMMON MISTAKE The single most damaging habit in the industry: updating the schedule's percent complete from duration elapsed ("the activity is 10 days into its 20-day duration, so it's 50%"). Time passing is not work completed. If your progress system cannot tell the difference, you do not have a progress system — you have a calendar.

Why accurate progress reporting matters

The consequences of measurement quality flow in both directions — bad and good:

When progress is unreliable	When progress is trustworthy
Forecasts are wrong; problems surface too late to fix cheaply	Early warning while recovery options are still cheap
Payment disputes and claim wars	Clean progress claims, faster certification and cash flow
Management debates whose numbers are right	Management debates what action to take
EOT claims collapse under scrutiny of the progress record	Contemporaneous records support entitlement
Site and office maintain parallel "truths"	Single source of truth; energy goes into the work

There is also a quieter, career-level consequence. The planner or project controls engineer whose numbers survive audit after audit becomes the person management calls before decisions — not after them.

Owner vs Contractor perspective

Progress measurement serves both parties, but they read it differently, and a professional must understand both lenses:

The contractor measures progress to drive execution (where are we behind?), to support invoicing (progress claims), to feed productivity analysis (earned vs expended hours), and to protect contractual entitlements (a contemporaneous record of delay events and their effects).

The owner measures progress to verify payment is earned before it is certified, to independently test the contractor's forecast, to detect trouble early enough to intervene, and to hold the contractor accountable to the approved baseline.

The healthiest projects establish one measurement system, defined in an approved Progress Measurement Procedure, that both parties trust — with the owner's verification rights built in. When each party runs its own system, the monthly progress meeting becomes an argument about arithmetic instead of a conversation about the project.

★ **INDUSTRY BEST PRACTICE** Agree the entire measurement framework — weightages, rules of credit, cut-off calendar, verification method, reporting formats — before the first progress claim, and lock it in a client-approved Progress Measurement Procedure. Every hour invested in that document repays itself many times over the project's life.

How this book is organised

Chapters 1–4 build the conceptual foundation: the control cycle, the measurement methods, and how to construct a weighting system. Chapters 5–8 are the machinery: implementing weighted progress in Primavera P6, earned value management with fully worked examples, S-curve analysis, and industry-standard progress rules. Chapters 9–12 apply the machinery to each project phase — engineering, procurement, construction — ending in a ready-to-use measurement matrix. Chapters 13–17 bring the practice up to date: Power BI dashboards, AI applications, EPC governance lessons, case studies, and the template library that accompanies this book.

The principles apply to any scheduling platform — Primavera P6, Oracle Primavera Cloud, Microsoft Project, Asta Powerproject — and to any industry that executes projects. Where software steps are shown, we use P6 as the reference tool because it remains the de facto standard in EPC and heavy industry.

Chapter 2: The Project Control Cycle

Progress measurement does not exist in isolation. It is one station in a repeating cycle that runs from the first plan to the final handover — and understanding where it sits in that cycle explains both why it matters and what it must produce for the stations downstream.

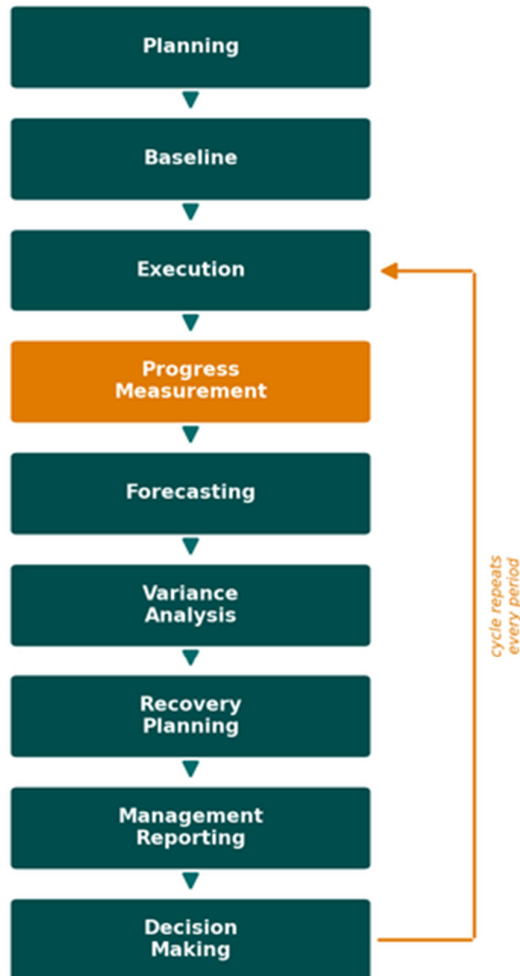


Figure: Project Control Cycle — nine-stage loop: Planning → Baseline → Execution → Progress Measurement → Forecasting → Variance Analysis → Recovery Planning → Management Reporting → Decision Making → (back to Execution)

The nine stations

1. Planning. Scope is decomposed (WBS), activities defined, logic built, durations estimated, resources loaded. The quality ceiling of every later station is set here — you cannot measure progress meaningfully against a plan that never reflected how the work would actually be done.
2. Baseline. The approved plan is frozen as the contractual reference. From this moment, the baseline is the yardstick; it changes only through formal, documented re-baselining.
3. Execution. The work happens. From a controls perspective, execution generates the raw data — quantities installed, documents issued, milestones achieved, hours expended.
4. Progress Measurement. The subject of this book: converting raw execution data into an objective statement of completion, using agreed weightages and rules of credit.
5. Forecasting. Measured progress plus remaining-work assessment produces the forecast — new completion dates, estimate at completion. A forecast not anchored in measured progress is an opinion.
6. Variance Analysis. Forecast and actuals are compared against the baseline: schedule variance, cost variance, SPI, CPI, milestone slippage. Variance analysis answers "how far off are we, and why?"
7. Recovery Planning. Where variance is unacceptable, options are developed: re-sequencing, acceleration, scope or resource decisions — each priced in time, cost and risk.
8. Management Reporting. Findings and options are communicated: progress report, S-curves, dashboards, narrative. Reporting is not decoration; it is the interface between analysis and authority.
9. Decision Making. Management acts — approves recovery, reallocates resources, escalates to the client. Decisions change execution, and the cycle turns again.

The cycle's clock speed

The cycle runs at different speeds simultaneously: weekly for site-level control (look-ahead, quantity tracking), monthly for formal reporting and progress claims, and per-shift on shutdowns and turnarounds where a day of delay costs millions. Whatever the speed, the sequence is the same — and a weakness at any station degrades every station after it.

 **APEX TIP** When diagnosing a project controls function, find the weakest station. Beautiful dashboards (station 8) built on estimated progress (station 4) are the most common — and most dangerous — configuration in the industry: confident reporting of unreliable numbers. Fix the measurement station first; everything downstream inherits the repair.

Where the money flows through the cycle

Progress measurement is also the commercial artery of the project. On most EPC contracts, measured progress drives the monthly progress claim; certified progress drives the payment certificate; and payment drives the contractor's cash flow. This is why measurement integrity is not

merely a technical virtue — a percent of progress on a RM500 million contract is RM5 million of cash flow. Both parties have every incentive to get the system right, and every incentive to argue when it is vague.

 **AI INSIGHT** Modern AI assistants can accelerate almost every station of this cycle — drafting the progress narrative from your variance data, sanity-checking weightage tables, generating recovery option lists for brainstorming. What they cannot do is stand behind the numbers. Throughout this book, AI Insight boxes show where the technology genuinely helps — and where professional judgement must remain firmly human.

Chapter 3: Methods of Measuring Progress

There is no single correct way to measure progress — there is a correct way for each type of work. The professional's skill lies in choosing the right method per work type and combining them into one coherent system. This chapter examines the five families of measurement methods, their strengths, their failure modes, and where each belongs.

Time-based measurement

Progress is inferred from schedule duration: an activity 12 days into a 20-day duration is deemed 60% complete.

Advantages

Zero data collection effort — the schedule calculates it automatically.

Acceptable for activities whose output genuinely tracks time.

Disadvantages

Measures the calendar, not the work. An activity can consume 100% of its duration and produce 40% of its output.

Invites passive updating — schedules that "update themselves" hide problems until they are unrecoverable.

Best applications

Level-of-effort and time-driven activities only: site supervision, scaffolding rental, temporary facilities, project management support. Never for physical deliverable work.

Quantity-based measurement

Progress = quantity installed ÷ total quantity, measured in the commodity's natural unit. The gold standard for repetitive physical work:

Commodity	Unit	Typical evidence
Concrete	m ³	Pour cards, delivery orders
Structural steel	tons	Erection reports, survey acceptance
Cable	meters	Cable pulling schedule/register
Pipe	inch-dia (welding), meters (erection)	Weld log, test packs
Equipment	units/tagged items	Installation reports
Painting/coating	m ²	Inspection reports (DFT records)
Earthwork	m ³	Survey volumes

Advantages

Objective and auditable — quantities can be physically verified.

Doubles as the input for productivity analysis and forecasting remaining work.

Disadvantages

Requires disciplined quantity tracking from day one — registers, take-offs, installed records.

Quantity growth (design development increasing the total) silently deflates reported progress unless totals are formally managed.

⚠ COMMON MISTAKE Dividing installed quantity by the ORIGINAL take-off after the design has grown. If installed cable is 40,000 m and the take-off grew from 60,000 m to 80,000 m, your progress is 50%, not 67%. Always measure against the current controlled total, and record every change to it.

Resource-based measurement

Progress is inferred from resource consumption: labour hours expended, equipment hours run, plant utilisation.

Advantages

Data is usually already collected (timesheets, plant records).

Useful for cost control and productivity context.

Disadvantages

Effort is not output — the fundamental flaw. Expended hours measure what the work cost, not how much got done.

Rewards inefficiency if misused: the slower the crew, the more "progress".

Best applications

Never as the primary physical progress measure. Use expended hours **ALONGSIDE** earned hours to compute productivity factor (Chapter 6) — that comparison is where resource data becomes powerful.

Cost-based measurement

Progress = cost incurred ÷ budget (or committed cost ÷ total commitment). Common in owner organisations and cost-reimbursable environments.

Advantages

Aligns with financial reporting and cash flow management.

Simple to produce from the accounting system.

Disadvantages

Money spent is not work done — an advance payment or early equipment PO moves cost dramatically with zero site progress.

Accounting lag (invoices arrive 30–90 days after work) makes it a rear-view mirror.

Best applications

Financial progress reporting and cash flow analysis — always presented alongside, never instead of, physical progress. When management sees "cost progress 55%, physical progress 43%" side by side, the 12-point gap IS the message.

Weightage-based measurement

The integrating method: every scope element receives a weight proportional to its share of the total (usually by budgeted manhours or cost), progress of each element is measured by the method best suited to it, and the weighted results roll up into one number. Virtually every serious EPC progress

system is weightage-based at the top, with quantity, milestone and document-based measurement underneath.

How the hierarchy works

1. Project level: Engineering 15% / Procurement 30% / Construction 48% / Commissioning 7% (weights from budgeted hours or value).
2. Within construction: disciplines weighted by budgeted hours (Civil 18%, Structural 18%, Mechanical 12%, Piping 30%, E&I 22%).
3. Within each discipline: work packages or commodity groups, measured by quantity or rules of credit.
4. Roll-up: each level's progress = Σ (child weight $\times$ child progress).

Common mistakes

Weights that don't sum to 100% after scope changes.

Weighting by contract value where value is distorted by commercial pricing (front-loaded rates flow straight into front-loaded progress).

Changing weights mid-project without approval or an audit trail.

Best practices

Derive weights from budgeted manhours — the most neutral proxy for work content.

Freeze weights in the approved procedure; change them only through formal revision with client sign-off.

Keep the weight register (Chapter 17 template) under document control like any other controlled deliverable.

Choosing: a decision guide

If the work is...	Measure by...
Repetitive, physical, countable	Quantity
Discrete deliverables (documents, POs)	Milestone / rules of credit
Continuous support (supervision, temp facilities)	Time / level of effort
Short and not worth detailed tracking	Simple progress rules (Ch. 8)
The whole project	Weightage roll-up of all the above

Chapter 4: Building a Weighting System

The weighting system is the skeleton of the entire progress measurement framework. Built well, it makes every later measurement mechanical. Built badly, it embeds distortion that no amount of diligent measuring can correct. This chapter walks through constructing one for a typical EPC project.

Principle: weight follows work content

The weight of any element should reflect its share of the total work content. The best available proxy is budgeted manhours; budgeted cost is the alternative where hours are not estimated (e.g., procurement packages, subcontracted lump sums). What weight must NEVER follow is convenience, negotiating position, or the desire to show early progress.

Step 1 — Phase-level weights

Start at the top. For a typical EPCC plant project, budgeted hours might produce:

Phase	Budgeted manhours	Weight
Engineering	180,000	12.0%
Procurement (management effort + value proxy)	300,000	20.0%
Construction	870,000	58.0%
Testing & Commissioning	120,000	8.0%
Handover & Close-out	30,000	2.0%
TOTAL	1,500,000	100.0%

Procurement deserves a note: its direct manhours (buyers, expeditors) are small, but the phase controls enormous value and schedule risk. Many systems weight procurement by material value converted through a norm, or agree a fixed phase weight with the client. Either is defensible — what matters is that the basis is explicit and approved.

Step 2 — Discipline weights within each phase

Within Construction (58% of the project), budgeted direct hours distribute across disciplines:

Discipline	Hours	Weight in phase	Weight in project
Civil & Earthworks	156,600	18.0%	10.4%
Structural Steel	156,600	18.0%	10.4%
Mechanical / Equipment	104,400	12.0%	7.0%
Piping	261,000	30.0%	16.8%
Electrical & Instrumentation	191,400	22.0%	12.8%
TOTAL Construction	870,000	100.0%	58.0%

The right-hand column — weight in project — is the number that matters in the monthly report: complete ALL piping and the project moves 16.8 points.

Step 3 — Work package and commodity weights

Within each discipline, decompose to the level at which measurement will actually happen — work packages, areas, or commodity groups — and weight them the same way. Stop decomposing when the element below would be measured identically to its parent; excess granularity adds administration without adding accuracy.

Step 4 — Choose the measurement basis per element

For each bottom-level element, assign the method (from Chapter 3) and the rule (from Chapter 8). This pairing — element, weight, basis, rule — is one row of your Progress Measurement Matrix (Chapter 12) and one row of the Weighting Register template (Chapter 17).

Worked micro-example

A piping work package for one area, budgeted at 42,000 hours (weight computed accordingly), decomposes as:

Element	Basis	Weight in package
Spool fabrication	inch-dia welded (quantity)	30%

Erection	inch-dia erected (quantity)	35%
Welding at site	inch-dia welded (quantity)	20%
Supports	units installed	5%
Testing (hydrotest)	test packs accepted	8%
Reinstatement & punch	punch items closed	2%

If fabrication is 80% done, erection 45%, site welding 30%, supports 50%, testing 0%: package progress = $0.30 \times 80 + 0.35 \times 45 + 0.20 \times 30 + 0.05 \times 50 + 0.08 \times 0 + 0.02 \times 0 = 24 + 15.75 + 6 + 2.5 = 48.25\%$.

 **APEX TIP** Notice testing carries 8% even though its manhours are modest. Weight can legitimately reflect criticality as well as hours where the client agrees — untested pipe is unusable pipe. If you adjust weights away from pure hours, document the rationale in the procedure. Transparency is what makes adjustments defensible.

Handling scope change

Every approved variation changes work content, so it changes weights. The disciplined process: recompute affected weights from revised budgets, issue a revision of the weighting register, obtain approval, and — critically — restate the baseline progress curve if the change is material. Historical actuals are never restated; they were true when measured.

 **COMMON MISTAKE** "Absorbing" scope growth without reweighting. The new work exists physically but not in the denominator, so reported progress drifts upward from reality — until the gap is discovered at 95% reported / 85% real, in the most painful month of the project to discover anything.

Case study: the weighting decision that saved a claim

On a Malaysian brownfield gas project, the contractor's planner insisted — against commercial pressure — on weighting tie-in works by realistic hours including permit and shutdown constraints, not by the small direct quantities involved. Tie-ins came to 6% of the project instead of 1.5%. When client shutdown windows slipped and tie-ins stalled for eleven weeks, the progress impact was visible at project level, the S-curve flattened unmistakably, and the contemporaneous progress record substantiated an extension of time claim that a 1.5% weighting would have rendered

invisible. Weighting is not just measurement mechanics; it determines what your records can prove later.

Chapter 5: Weighted Progress in Primavera P6

This chapter turns the weighting system of Chapter 4 into a working implementation in Primavera P6. The same logic transfers to Oracle Primavera Cloud, Microsoft Project or Asta Powerproject — only the buttons differ.

The implementation choice

There are two proven architectures, and choosing between them is the first decision:

- 1. Inside P6:** load budgeted units (hours) onto activities; P6 computes weighted progress automatically as earned units roll up through the WBS. Best when the schedule is detailed enough to BE the measurement system.
- 2. Outside P6, and connected:** maintain detailed measurement (deliverable registers, quantity trackers) in Excel or a database; compute physical % per activity there; enter it into P6 during the update. Best when measurement detail is finer than schedule detail — the norm on large EPC projects.

Most mature EPC systems are hybrid: quantity and document detail lives outside, P6 carries activity-level physical % and produces the time-phased roll-up.

Setting up P6 for weighted progress

Step 1 — Load budgeted units

Assign resources with Budgeted Units to every activity (crew size × duration × hours/day, or estimate-derived hours). An activity's share of total budgeted units IS its weight — no separate weight field needed.

Activity ID	Activity Name	Original Duration	Predecessors	Start	Finish	Budgeted Labor Units	Total Float
1222	EPC LEVEL III SCHEDULE	1222		23-Jul-14 A	26-Jun-19	22576996	402
1222	ZONE 0 - GENERAL	1222		22-Jul-14 A	26-Jun-19	1936426	402
1176	ZONE 1 - REFINERY	1176		04-Aug-14 A	30-Apr-19	19324063	45
1129	UNIT 1120 - SATURATED GAS PLANT	1129		04-Aug-14 A	20-Feb-19	1532161	2
729	HOME OFFICE SERVICES	729		04-Aug-14 A	05-Jan-18	65422	281
635	MANUFACTURING	635		11-May-15 A	03-Mar-18	0	241
690	CONSTRUCTION	690		01-Aug-15 A	06-Jun-18	1207762	26
107	Piping	107		01-Aug-15 A	20-Jan-16 A	3686	
560	Civil	560		14-Oct-15 A	10-Mar-18	341593	60
495	Steel Structure	495		15-Jan-16 A	03-Feb-18	77482	78
451	Mechanical	451		21-Mar-16 A	07-Apr-18	54128	38
487	Piping	487		18-Apr-16 A	06-Jun-18	520367	26
268	Piping Prefabrication	268		18-Apr-16 A	13-Dec-17	123143	7
268	C31120P1P000010 Piping Prefabrication Rack Unit 1120	268	E 335131150010401C	18-Apr-16 A	06-Nov-17	31240	7
256	C31120P1P000020 Piping Prefabrication Big Bore Unit 1120	256	E 335131150010401C	21-Apr-16 A	13-Dec-17	91903	7
253	Piping Erection (Big Bore)	253		05-Oct-16 A	10-Feb-18	146600	8
155	C31120P1B000020 Area A - BB Piping Erection Unit 1120	155	M000131F1077200G	05-Oct-16 A	10-Feb-18	7330	8
195	C31120P1B000030 Area B - BB Piping Erection Unit 1120	195	M000131F1077200G	04-Feb-17 A	26-Jan-18	32252	7
155	C31120P1B000040 Area C - BB Piping Erection Unit 1120	155	M000131F1077200G	03-Jan-17 A	13-Dec-17	26388	37
145	C31120P1B000050 Area D - BB Piping Erection Unit 1120	145	M000131F1077200G	04-Jan-17 A	23-Nov-17	14660	29
170	C31120P1B000060 Area E - BB Piping Erection Unit 1120	170	M000131F1077200G	02-Jan-17 A	08-Jan-18	32252	12
187	C31120P1B000070 Area F - BB Piping Erection Unit 1120	187	M000131F1077200G	04-Jan-17 A	29-Jan-18	14660	9
170	C31120P1B000080 Area G - BB Piping Erection Unit 1120	170	M000131F1077200G	16-Jan-17 A	08-Jan-18	7330	13
90	C31120P1B000090 Area H - BB Piping Erection Unit 1120	90	M000131F1077200G	22-Dec-16 A	06-Sep-17	11728	31
166	Piping Erection (Small Bore)	166		05-Jul-17	20-Feb-18	115344	8

Step 2 — Set % Complete Type

Set activities to Physical % Complete (Activity Details > General tab). Set it as the project default before creating activities. Duration % is the enemy identified in Chapter 1; Units % suits only activities where earned hours are directly recorded.

Step 3 — Configure Earned Value settings

In Admin Preferences (or per-WBS in Project settings), set Earned Value calculation to use Physical % against Budgeted Units. Performance % Complete drives the earned value fields — align it with Physical %.

The screenshot shows a software interface with a tabbed menu at the top: General, Notebook, Budget Log, Spending Plan, Budget Summary, WBS Milestones, WPs & Docs, and Earned Value. The 'Earned Value' tab is selected. Below the tabs are two main sections:

- Technique for computing performance percent complete**
 - ☒ Activity percent complete
 - ☐ Use resource curves / future period buckets
 - ☐ WBS Milestones percent complete
 - ☐ 0/100
 - ☐ 50/50
 - ☐ Custom percent complete: 6

- Technique for computing Estimate to Complete (ETC)**
- ☒ ETC = remaining cost for activity or
- ETC = $PF * (\text{Budget at Completion} - \text{Earned Value})$, where:
 - ☐ PF = 1
 - ☐ PF = 1 / Cost Performance Index
 - ☐ PF = 1 / (Cost Performance Index * Schedule Performance Index)
 - ☐ PF = 0.88

Step 4 — Verify the roll-up

Group by WBS and display the Performance % Complete column: each WBS band now shows the weighted progress of its children — automatically weighted by budgeted units. Spot-check one band by hand calculation before trusting the machine. Always.

The monthly update workflow

1. Cut-off: freeze site data on the agreed date.
2. Collect: quantity trackers, document register status, milestone confirmations — from the measurement systems, not from memory.
3. Compute: physical % per activity in the measurement workbook.
4. Enter: physical % + actual dates + remaining durations into P6.
5. Schedule (F9) with the new data date.
6. Extract: earned/planned/actual values for reporting (next section).

7. Reconcile: P6 roll-up vs measurement workbook total — they must match to the decimal. Investigate any gap; it is always a data error somewhere.

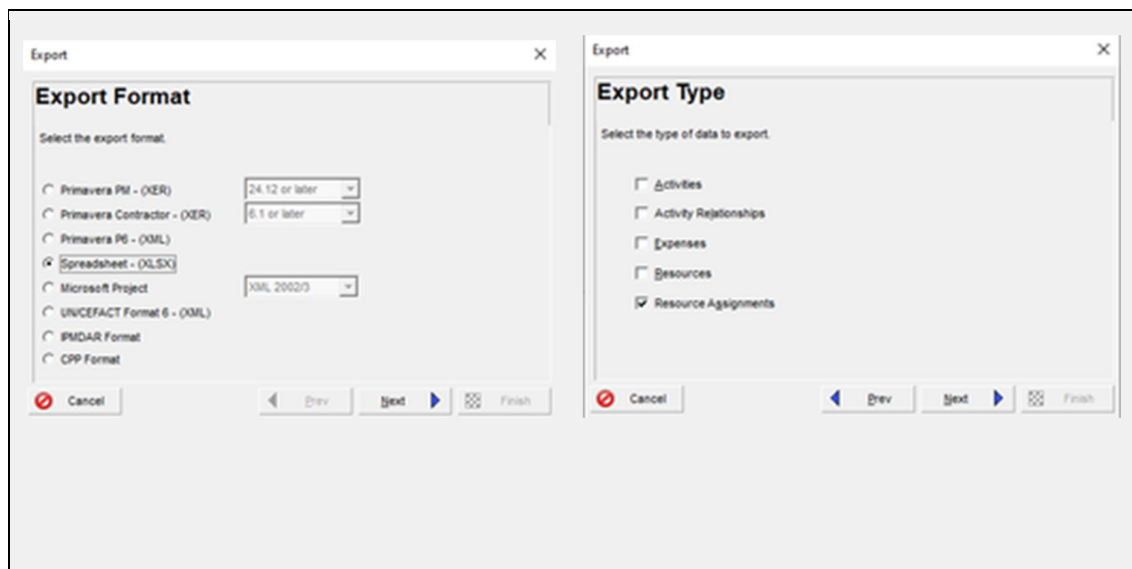
APEX TIP Automate step 4 where volume justifies it: P6's Import from Excel (File > Import > Spreadsheet) updates Physical % and dates against Activity ID. Export the update template first (File > Export), have the site fill exactly that format, and import. Fewer typing errors, faster cycle, and a file audit trail of every monthly submission.

Getting data out: export and reporting

Time-phased spreads: Resource Assignments view with monthly Budgeted/Actual/Remaining Units columns → export to Excel → feeds S-curves (Chapter 7) and dashboards (Chapter 13).

Progress by WBS: Activities view grouped by WBS with Performance %, Schedule %, Physical % columns → the backbone of the monthly report's progress table.

Earned value fields: Planned Value, Earned Value, Actual Cost columns (labour units or cost) per activity or WBS band.



What P6 will not do for you

P6 will not verify quantities, chase the site for evidence, notice that a weight is front-loaded, or explain a variance to management. It is an exceptional calculation and time-phasing engine sitting on top of whatever measurement discipline you feed it. The system is the procedure, the registers, and the habits — P6 is the calculator.

Chapter 6: Earned Value Management

Earned Value Management is the integration layer: it connects scope, schedule and cost into a single framework so that one set of metrics answers both "are we late?" and "are we over budget?" — and, uniquely, "how late and how much, by the end?" This chapter builds the full EVM toolkit with one continuous worked example.

The three fundamental values

Everything in EVM derives from three numbers, each measured at the same status date:

PV — Planned Value (BCWS): the budgeted value of work SCHEDULED to be complete by now. From the baseline.

EV — Earned Value (BCWP): the budgeted value of work ACTUALLY complete by now. Budget × measured physical progress.

AC — Actual Cost (ACWP): what the completed work actually COST. From the cost system.

Note the discipline hidden in EV: it prices actual work at BUDGETED rates. EV isolates volume of work done from the price paid to do it — that separation is what makes the schedule and cost signals independent.

The worked example

Project: process plant upgrade. BAC (Budget at Completion) = RM 20,000,000. Status: end of Month 8 of a planned 14 months.

Value	Amount (RM)	Source
PV	11,000,000	Baseline: 55% of work scheduled done by Month 8
EV	9,200,000	Measured physical progress 46% × BAC
AC	10,350,000	Cost system

Variances: how far off are we?

Schedule Variance: $SV = EV - PV = 9.2M - 11.0M = -RM\ 1,800,000$. We have earned RM1.8M less than planned — behind schedule.

Cost Variance: $CV = EV - AC = 9.2M - 10.35M = -RM\ 1,150,000$. The work done has cost RM1.15M more than its budget — over cost.

Indices: how efficient are we?

$SPI = EV \div PV = 9.2 \div 11.0 = 0.84$. We are progressing at 84% of the planned rate.

$CPI = EV \div AC = 9.2 \div 10.35 = 0.89$. Every ringgit spent buys RM0.89 of budgeted work.

Index	> 1.0	= 1.0	< 1.0
SPI	Ahead of schedule	On schedule	Behind schedule
CPI	Under budget	On budget	Over budget

Forecasts: where will we end up?

EAC (Estimate at Completion), performance-based: $EAC = BAC \div CPI = 20.0M \div 0.889 = RM\ 22,500,000$.

EAC, remaining-work-at-budget (optimistic — assumes problems won't recur): $EAC = AC + (BAC - EV) = 10.35M + 10.8M = RM\ 21,150,000$.

EAC, combined index (schedule pressure inflates cost): $EAC = AC + (BAC - EV) \div (CPI \times SPI) = 10.35M + 10.8M \div 0.747 = RM\ 24,800,000$.

ETC (Estimate to Complete) = $EAC - AC = 22.5M - 10.35M = RM\ 12,150,000$ (using the CPI-based EAC).

VAC (Variance at Completion) = $BAC - EAC = 20.0M - 22.5M = -RM\ 2,500,000$ forecast overrun.

The three EAC formulas bracket the outcome: RM21.2M if all problems stop today, RM22.5M if cost performance continues, RM24.8M if schedule pressure compounds it. Presenting the bracket — not one number — is the honest forecast.

TCPI: the reality check

To-Complete Performance Index asks: what CPI must the REMAINING work achieve to hit a target?

To hit BAC: $TCPI = (BAC - EV) \div (BAC - AC) = 10.8M \div 9.65M = 1.12$.

To hit the EAC of 22.5M: $TCPI = (BAC - EV) \div (EAC - AC) = 10.8M \div 12.15M = 0.89$.

Read that first line carefully: a team that has performed at CPI 0.89 for eight months must suddenly perform at 1.12 — 26% better — for six months to land on budget. When someone claims "we'll recover to budget", TCPI quantifies exactly how heroic that claim is.

★ **INDUSTRY BEST PRACTICE** Research across hundreds of projects (the basis of US DoD EVM practice) shows CPI stabilises remarkably early — rarely improving by more than a few points after the 20% completion mark. A project at CPI 0.89 at 46% complete will, absent structural change, finish near CPI 0.89. Treat performance-based EAC as the central estimate, not the pessimistic one.

Schedule variance in time: earned schedule

SV and SPI are denominated in money, which creates two known weaknesses: they converge to zero/1.0 at completion regardless of lateness, and they say nothing about the critical path. Earned Schedule converts the same data into time: find the date when the baseline planned the current EV — the horizontal gap to the status date is the schedule variance in weeks. In our example, PV reached RM9.2M in mid-Month 6, so the project is roughly 7 weeks behind. Always pair EVM metrics with critical-path analysis: EVM tells you the size of the problem; the network tells you whether it hits the completion date.

⚠ **COMMON MISTAKE** Reporting SPI on a project that is 92% complete and calling the schedule "recovering" because SPI rose from 0.84 to 0.95. Late-project SPI always drifts toward 1.0 — it is arithmetic, not recovery. Use earned schedule or milestone variance for late-stage reporting.

Manhours EVM: the contractor's daily version

Everything above works identically in manhours — often more practically for contractors, since hours avoid commercial rates: earned hours = budget hours × physical %; Productivity Factor (PF) = earned ÷ expended hours (the manhours CPI). The Chapter 17 Earned Value Worksheet implements both currencies side by side.

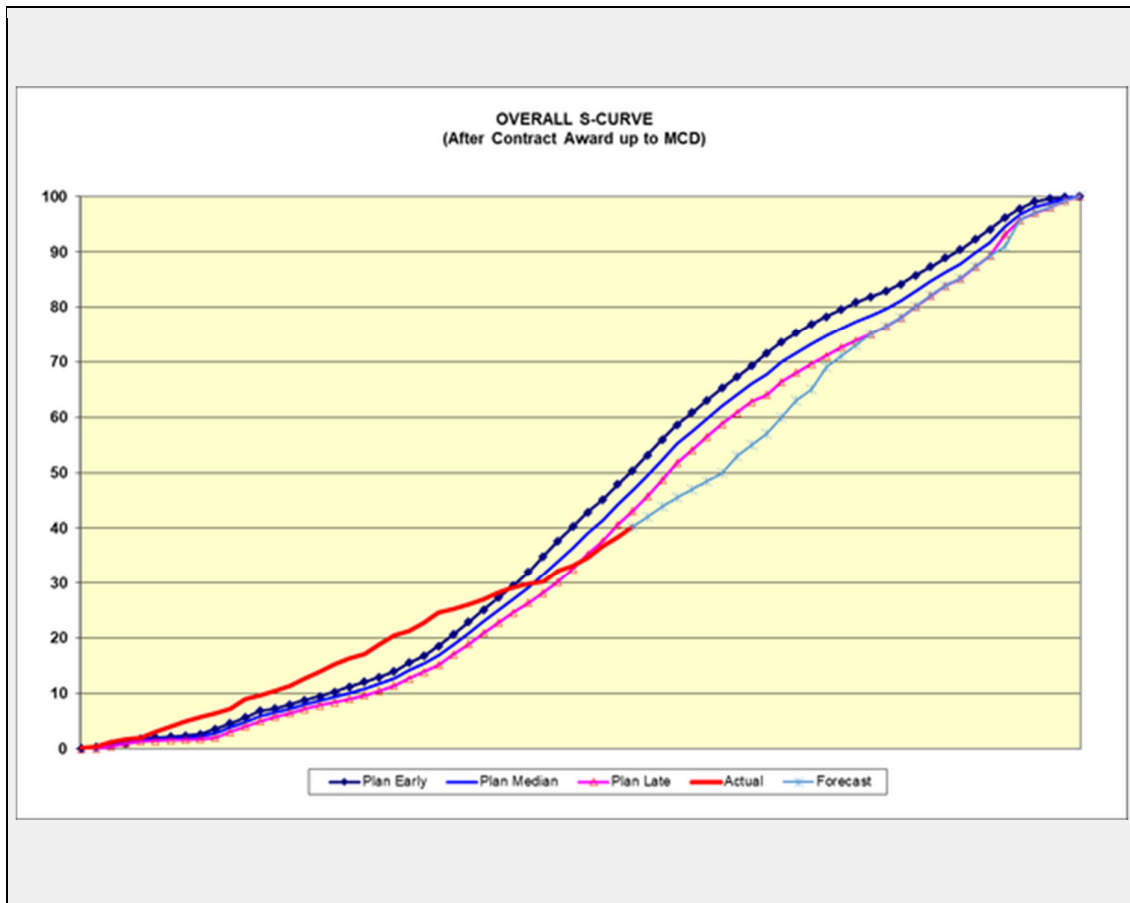
Chapter 7: S-Curve Analysis

The S-curve is the most recognised artefact in project controls — cumulative progress (or value, or hours) plotted against time, its characteristic shape produced by the slow start, intense middle and long tail of every real project. This chapter covers the four curves that belong on the chart, how to read them, and how to use them in recovery and executive reporting.

The four curves

1. Planned (Baseline) curve — cumulative PV from the approved baseline, time-phased through the schedule's resource/cost spread. Fixed unless formally re-baselined.

2. Actual (Earned) curve — cumulative EV at each cut-off: the measured physical progress record. It exists only up to the data date. (Where cost is plotted, a separate Actual Cost curve shows AC — the gap between earned and actual-cost curves is your cost variance made visible.)
3. Forecast curve — the remaining work time-phased from the current schedule, extending the earned curve from the data date to forecast completion.
4. Late curve — the baseline spread on late dates. Early + late curves form the "banana"; an earned curve below the late curve means the completion date is no longer achievable without recovery.



Reading the chart: the standard patterns

Pattern	Reading	Response
Earned parallel below plan	Constant lag; rate matches plan	Recoverable with modest acceleration — or accept and re-forecast
Gap widening month on month	Progress rate below planned rate	Structural problem: resources, productivity, or front availability. Rate must change
Earned curve flattening	Work (or reporting) has stalled	Investigate immediately — both causes are serious
Earned above plan early, flattening later	Easy work done first / front-loaded weights	Audit weightages; expect a hard tail
Sudden step up	Bulk credit taken (milestone cluster, data correction)	Verify evidence before celebrating
Forecast curve steeper than any historical actual rate	Recovery assumed but not resourced	Challenge: what specifically changed?

That last row deserves emphasis. The forecast curve's slope is a promise about future monthly production. If the project has never achieved 6% in a month, a forecast requiring five consecutive 8% months is not a plan — it is hope drawn in Excel.

 **APEX TIP** Plot the required rate explicitly: $(100\% - \text{current progress}) \div \text{months remaining}$, as a monthly figure, next to the best month ever achieved. When required rate exceeds best-ever rate, that single comparison ends more arguments than any variance table.

Recovery on the S-curve

A credible recovery plan appears on the chart as a new forecast curve with three properties: it starts from the actual position (not from where we wish we were), its slope is justified by named actions (added crews, extended hours, re-sequenced fronts, released constraints), and it shows the cost of

those actions alongside. Management approves recovery curves; it should never be shown "recovery" that is merely a redrawn line.

The executive dashboard views

For senior audiences, the S-curve carries the headline, supported by a compact metrics block:

The chart: plan, earned, forecast curves; data date line; contract completion and forecast completion flagged.

KPI cards: overall % (plan vs actual), SPI, CPI, forecast completion date vs contract, float on critical path.

One paragraph: position → cause → action (the three-sentence narrative discipline from the Apex S-curve guide).

An executive should extract the project's condition in thirty seconds and the story in two minutes. Everything else is appendix.



AI INSIGHT Feed the monthly S-curve data table to an AI assistant and ask for the three-sentence narrative draft — position, cause, action. It drafts in seconds what takes an hour to write from scratch. Then correct it: the AI knows the arithmetic; it does not know that the piping delay is really a paint-shop dispute. The narrative's facts remain the planner's responsibility.

Chapter 8: Progress Rules

Not every activity justifies quantity tracking or a detailed rules-of-credit table. Progress rules are the industry's standard shorthand techniques — each an agreed convention for when credit is taken. Choosing the right rule per activity type keeps the system accurate where it matters and cheap where it doesn't.

The standard rules

0/100

No credit until the activity is fully complete. The conservative rule for short activities — nothing is earned until everything is verifiable. Best for: short-duration discrete tasks (a lift, a tie-in, a final document), anything within one reporting period.

50/50

Half credit at start, half at completion. Rewards mobilisation on activities spanning two or three periods, while capping the exposure of unfinished work. Best for: short activities crossing one cut-off.

20/80 and 25/75

Small credit at start, bulk at completion — the middle ground where starting is easy but finishing is the work. Common for installation tasks where "started" is trivially achieved. Best for: activities where the temptation to claim early progress needs structural restraint.

Milestone (weighted milestones)

Credit at defined intermediate events, each with a fixed percentage — the rules-of-credit approach of Chapter 4's examples (fabricated 40 / erected 30 / welded 20 / tested 10). The workhorse rule for anything with verifiable intermediate states. Best for: equipment installation, document production, package work.

Physical % (measured)

Continuous measurement against quantity — $\text{installed} \div \text{total}$. The most accurate and most demanding rule. Best for: high-volume commodities (concrete, cable, welding, coating).

Incremental (units completed)

$\text{Progress} = \text{completed units} \div \text{total units}$, where each unit is itself 0/100. A hybrid: continuous at package level, binary at unit level. Best for: repetitive identical items — piles driven, rooms completed, loops tested.

Level of Effort (LOE)

Progress follows elapsed time, deliberately. Honest for work that has no output of its own. Best for: supervision, site services, scaffolding maintenance, project controls itself. LOE activities should carry minimal weight — they exist to spread indirect hours, not to influence the progress number.

Selection guide

Activity character	Rule
Short, discrete, within one period	0/100
Short, spans a cut-off	50/50
Easy to start, hard to finish	20/80 or 25/75
Verifiable intermediate stages	Weighted milestones
Countable high-volume commodity	Physical % (quantity)
Many identical units	Incremental
Time-driven support	LOE

The rules govern claims, not reality

A subtlety worth stating plainly: progress rules are conventions about when credit may be CLAIMED, deliberately simplified from physical reality. A 50/50 activity that is physically 70% done still reports 50%. That conservatism is the point — the rules trade fine accuracy on small activities for auditability and speed across thousands of them. Precision belongs where the hours are: on the heavily weighted commodities measured by quantity.

 **COMMON MISTAKE** Letting individual activities negotiate their rule after work starts ("this one's really 80% even though it's 50/50..."). The moment rules bend per-activity, the system is estimation with extra steps. Rules are fixed in the procedure, per activity type, before execution — exceptions require the same formal revision as a weight change.

★ **INDUSTRY BEST PRACTICE** In the Progress Measurement Procedure, publish a one-page table mapping every activity type to its rule (the Chapter 12 matrix). During progress verification, the client checks claimed credit against that table mechanically. When both parties can audit progress with a lookup table, monthly certification stops being a negotiation.

Chapter 9: Engineering Progress Measurement

Engineering produces documents, not concrete — so engineering progress is measured through the deliverables register: every drawing, calculation, specification, datasheet and report, each weighted by budgeted hours and earning credit as it advances through defined issue states.

The deliverable lifecycle and standard credit steps

A typical credit ladder for a design deliverable:

Status	Credit	Cumulative	Evidence
Started / in progress	10%	10%	Register status
IDC — inter-discipline check complete	20%	30%	IDC record
IFR — Issued for Review	20%	50%	Transmittal
IFA — Issued for Approval (comments incorporated)	20%	70%	Transmittal
IFC — Issued for Construction	30%	100%	Transmittal

Percentages vary by company and contract — some ladders credit 60% at IFR for calculation-heavy documents; some withhold 5% until as-built. What matters is one agreed ladder per document class, applied without exception.

Weighting the register

Each deliverable's weight derives from its budgeted hours: a 3D-model-extracted piping isometric might carry 6 hours; a process simulation report 400. Engineering progress = Σ (deliverable weight ×

ladder credit). Discipline and area roll-ups come free if the register carries discipline/area attributes — and they should.

The revision cycle problem

Documents cycle: IFR attracts comments, revisions consume hours, sometimes a document returns to an earlier state. Standard treatments:

Progress never goes backwards for routine comment incorporation — the ladder already reserves credit (the IFA step) for that work.

Major recycles (client scope change, failed review requiring redesign) are handled as scope change: hours added to the budget, weights revised, and the cause documented — that record is claim evidence.

 **COMMON MISTAKE** Crediting 90% of engineering while vendor data is missing. Deliverables dependent on vendor information (foundations awaiting vendor loads, E&I awaiting certified drawings) cannot genuinely pass IFC. Track "IFC-able except vendor data" as its own visible status — it converts a hidden engineering problem into a visible procurement one, which is where it belongs.

Vendor data as engineering scope

Vendor document review is real engineering work and belongs in the register: each vendor document expected, weighted, and credited through its own ladder (received → reviewed → returned → final). Projects that leave vendor data outside the register systematically overstate engineering progress in the middle third of the phase.

Chapter 10: Procurement Progress Measurement

Procurement progress is milestone-based: each purchase order or package advances through a standard chain of verifiable events. The measurement challenge is not the method — it is weighting packages sensibly and resisting the temptation to declare victory at PO award.

The standard milestone chain

A typical credit ladder per package (weights adjusted per package criticality):

Milestone	Credit	Cumulative	Evidence
MR — Material Requisition issued	5%	5%	Requisition transmittal
RFQ issued to bidders	5%	10%	RFQ record

Bid evaluation complete (TBE/CBE)	10%	20%	Approved evaluation
PO placed	10%	30%	Signed PO
Vendor drawings approved	10%	40%	Document register
Manufacturing complete	25%	65%	Vendor notification / inspection
Inspection & FAT passed	10%	75%	Inspection release note
Shipped / delivered to site	20%	95%	Delivery order / MRR
Customs cleared & received (imported items)	5%	100%	Receiving report

Note where the weight sits: 60% of the credit lies BETWEEN PO placement and site delivery — the manufacturing and logistics period that actually consumes the schedule. Ladders that credit 60% by PO award describe purchasing effort, not procurement reality, and go blind for the six months that matter most.

Weighting packages

Weight packages by value (the standard proxy) with criticality adjustments agreed in the procedure: a long-lead compressor at 8% of material value may deserve more than 8% weight because the schedule lives or dies by it. As always — any departure from the neutral basis is documented and approved.

Expediting: measurement's active twin

The procurement progress report is also the expediting priority list: packages behind their milestone plan, sorted by schedule criticality (link each package to its need date from the schedule).

Measurement that doesn't drive expediting is bookkeeping.

 **APEX TIP** Track "forecast delivery vs need date" as a column beside milestone progress. A package can be 65% on its ladder and perfectly on plan — or 65% and six weeks behind its need date. The ladder measures the package; the float column protects the project.

Chapter 11: Construction Progress Measurement

Construction carries most of the weight, most of the hours and most of the measurement effort. The method is quantity-based measurement per commodity (Chapter 3), organised by discipline, with rules of credit standardising the stages within each commodity.

Discipline by discipline

Civil & Earthworks

Excavation and backfill by m³ (survey); piling by count (incremental — each pile 0/100 or a short ladder: driven/cut-off/tested); concrete by m³ through a rebar–formwork–pour–cure ladder. Evidence: survey reports, pour cards, pile records.

Structural Steel

By tonnes through fabrication–erection–alignment–bolting/final connection stages, or by piece count for modular/light steel. Fabrication (often off-site) and erection tracked as separate weighted elements. Evidence: fabrication releases, erection reports, survey acceptance.

Mechanical / Equipment

Per tagged item through weighted milestones: foundation ready → set/aligned → grouted → internals/lube → cold checks complete. Heavy lifts may deserve their own milestone. Evidence: installation reports, alignment records.

Piping

The classic inch-dia system: fabrication welding and field welding by inch-dia welded (weld log is the source of truth); erection by inch-dia or spool count; then supports, testing (by test pack), reinstatement. Piping is usually the largest single discipline weight — its measurement discipline sets the tone for the project.

Electrical

Cable by metres pulled (cable register), terminations by count, cable tray/ladder by metres, equipment (switchgear, transformers, panels) by weighted milestones, lighting by fixture count.

Instrumentation

Instruments installed by tag count; impulse/tubing by metres; terminations by count; loop checks by loop count (incremental — the natural unit of I&C completion).

Telecommunication

Backbone/fibre by metres, cabinets and access points by unit milestones, testing by segment/system count.

Testing & Commissioning

System-based, not area-based (the completion pivot): mechanical completion by subsystem (walkdown accepted, punch A cleared), pre-commissioning activities by defined test count (motor solo runs, flushing circuits, loop checks), commissioning by system milestone ladder through ready-for-start-up.

The quantity tracking backbone

Every discipline above presumes a controlled quantity database: take-off quantities by commodity by area, updated for design change, with installed quantities recorded weekly against evidence. This database (the Quantity Tracking template, Chapter 17) is the construction progress system — P6 merely receives its roll-up. Its disciplines: one owner, document-controlled totals, weekly capture, and evidence references on every entry.

★ **INDUSTRY BEST PRACTICE** Reconcile claimed quantities against physical verification monthly on a sample basis — walk down 5–10% of claimed items, alternating areas without announcement. Sampling keeps everyone honest at a fraction of the cost of full verification, and the client's own verifier finding your records accurate is worth more than any assurance letter.

⚠ **COMMON MISTAKE** Letting subcontractors self-report percentages instead of quantities. A subcontractor reports "cable pulling 70%" — against what total, measured how? Require quantities (metres pulled, per register), apply YOUR controlled totals and rules. Percentages are computed, never collected.

Chapter 12: The Progress Measurement Matrix

The matrix is the entire system on one page: every work type mapped to its unit, measurement basis, rule, evidence and source register. It belongs in the Progress Measurement Procedure as the master reference, and on the wall beside every progress engineer.

The master matrix

Work type	Unit	Basis	Rule	Evidence	Source register
Engineering deliverables	document	Milestone ladder	IFR/IFA/IFC ladder	Transmittals	Deliverables register
Vendor documents	document	Milestone ladder	Rec'd/Rev'd/Final	Doc register	VDR register
Procurement packages	PO	Milestone ladder	MR→delivery chain	PO milestones	Procurement register
Earthwork	m ³	Quantity	Physical %	Survey	Quantity tracker
Piling	no.	Incremental	Per-pile 0/100	Pile records	Pile register
Concrete	m ³	Quantity + ladder	Rebar/form/pour/cure	Pour cards	Quantity tracker
Structural steel	tonne	Quantity + ladder	Fab/erect/align/bolt	Erection reports	Steel register
Equipment	tag	Milestone ladder	Set/align/grout/check	Install reports	Equipment register
Pipe fabrication	inch-dia	Quantity	Physical %	Weld log	Weld register
Pipe erection	inch-dia	Quantity	Physical %	Erection records	Spool register

Pipe testing	test pack	Incremental	Per-pack 0/100	Test certificates	Test pack register
Cable	metre	Quantity	Physical %	Pull records	Cable register
Terminations	no.	Quantity	Physical %	Termination sheets	Cable register
Instruments	tag	Incremental	Per-tag ladder	Install/calibration	Instrument index
Loop checks	loop	Incremental	Per-loop 0/100	Loop folders	Loop register
Painting/coating	m ²	Quantity	Physical %	DFT reports	Coating register
Insulation	m ² / m	Quantity	Physical %	Inspection	Quantity tracker
Scaffolding	LOE	Time	LOE	—	(minimal weight)
Supervision/indirects	LOE	Time	LOE	—	(minimal weight)
Mechanical completion	subsystem	Incremental	Walkdown + punch A	MC certificates	Completion database
Commissioning	system	Milestone ladder	System ladder	Comm. records	Completion database

Using the matrix

Three rules make it work in practice. First, completeness: every activity in the schedule must map to exactly one row — anything unmapped will be estimated, and estimation spreads. Second, stability: the matrix changes only by procedure revision, never by monthly convenience. Third, visibility: the

matrix is shared with the client, the subcontractors and the site — a measurement system works precisely to the degree that everyone measuring knows the rules are the same for everyone.

 **APEX TIP** When a new work type appears mid-project (it always does), resist inventing its measurement in the monthly rush. Add a matrix row properly — unit, basis, rule, evidence, register — issue the revision, then measure. Thirty minutes of discipline versus a permanent soft spot in the system.

Chapter 13: The Power BI Progress Dashboard

The measurement system produces data; the dashboard makes it govern. Power BI (or any modern BI platform) turns the monthly and weekly progress cycle from document production into a living, drillable view. This chapter defines the two dashboards every project needs and the design discipline behind them.

Architecture: one source, two faces

Feed the dashboard from the same controlled sources that feed the report — the P6 exports, the quantity tracker, the deliverables and procurement registers — via Power Query, with the data date stamped on every page. One data model serves two audiences:

The executive dashboard (monthly)

Answers "what is the condition of the project?" in thirty seconds:

Headline S-curve: plan / earned / forecast, contract vs forecast completion flagged.

KPI cards: overall progress (plan vs actual), SPI, CPI, productivity factor, critical path float, forecast completion date.

Milestone trend chart: forecast date of each key milestone plotted across successive updates — the "is it still slipping?" chart, and the most underused visual in the industry.

Variance by area/discipline: one bar chart, sorted, worst first.

The narrative block: position, cause, action — three sentences, updated monthly, no more.

The working dashboard (weekly)

Answers "where do we act this week?":

Progress by discipline and area vs weekly plan (small multiples work well).

Quantity run-rates: installed per week vs required per week for each key commodity — the trade-level version of the required-rate test from Chapter 7.

Heat map: WBS area × discipline grid coloured by schedule variance — the fastest way to spot the corner of the project quietly falling behind.

Top 20 late activities by float impact; constraint list with owners and dates.

Productivity trends per discipline (earned vs expended hours, weekly).

Design discipline

Every visual answers one management question; if you cannot name the question, delete the visual.

Consistent semantics: red is always adverse, one decimal place for percentages.

Data date visible on every page; source and refresh time in the footer.

Drill path mirrors the WBS: project → area → discipline → work package. Three clicks from headline to the late cable register.



AI INSIGHT Power BI's built-in AI visuals earn their place here: Key Influencers can surface which areas/disciplines statistically drive schedule variance; Decomposition Tree lets management explore the variance hierarchy themselves; Smart Narrative drafts the caption text (edit it — Chapter 7 rules apply). Anomaly detection on weekly quantity series flags reporting errors (a claimed 300% week) before the client does.

Chapter 14: AI in Progress Measurement

AI is changing project controls faster than any technology since the CPM algorithm itself. Used well, it removes drudgery and sharpens analysis; used carelessly, it manufactures confident nonsense at scale. This chapter maps the current, realistic applications — and their guardrails.

Language models as analyst assistants (ChatGPT, Claude, Copilot)

Practical, high-value uses today:

Narrative drafting: variance tables in, structured progress narrative out — then corrected by someone who knows the ground truth.

Procedure and matrix drafting: generate the first draft of a Progress Measurement Procedure or rules-of-credit ladder for an unusual work type, then engineer it.

Schedule quality review: paste a schedule log or activity extract; ask for missing-logic patterns, suspicious constraints, duration outliers.

Data wrangling: transforming messy site returns into the standard import format — the assistant writes the Power Query / Python steps.

Learning accelerator: junior engineers interrogating concepts ("why is my SPI 0.9 but critical path on time?") with instant, patient explanation.

⚠ COMMON MISTAKE Pasting confidential project data — client names, commercial values, claim positions — into public AI tools without approval. Treat AI services like any other subprocessor: use enterprise deployments with data protection agreements, anonymise where possible, and follow the project's information security rules. One careless prompt can breach an NDA.

Power BI AI features

Covered in Chapter 13 — key influencers, decomposition trees, anomaly detection, smart narratives. Note the pattern in all of them: AI proposes, the analyst verifies, management consumes verified output only.

OCR and document intelligence

Progress evidence lives in paper-shaped documents: pour cards, weld logs, delivery orders, inspection reports. OCR-based document AI extracts structured data from these at scale — quantity, date, reference, signature presence — feeding the registers with far less manual entry. Mature tools reach high accuracy on standard forms; the discipline is a human verification sample, exactly like the physical quantity sampling of Chapter 11.

Drones, computer vision and reality capture

Drone photogrammetry produces orthophotos and 3D meshes of the site on a weekly cadence: earthwork volumes computed from imagery, laydown and structural erection visually confirmed against claims.

360° reality capture walks (helmet or tripod rigs) create timestamped visual records of every area — dispute-resolution gold, and increasingly the medium for remote progress verification.

Computer vision systems compare capture against the 4D model to flag installed/not-installed status for major elements. Current honest capability: strong on structure, equipment and bulk visibility; weak on small-bore, buried and internal work. Treat CV progress as corroboration, not measurement of record.

Digital twins and 4D integration

Linking the measurement system to the 3D/4D model turns progress into something management can SEE: elements coloured by status, the month's earned work highlighted, the forecast sequence animated. The measurement data underneath is unchanged — the twin is a presentation and verification layer, and a powerful one for client confidence.

Predictive analytics

Machine learning models trained on historical update data (own-company history is far more predictive than industry generic) can forecast: probable completion slippage from early-warning indicators, activities most likely to be reported inaccurately (audit targeting), and productivity

trajectories by discipline. Even simple statistical discipline — CPI stability research, reference-class forecasting of your forecast-vs-actual history — outperforms optimism. Start there before buying "AI forecasting".

★ **INDUSTRY BEST PRACTICE** Adopt AI tools in the order of least risk: (1) drafting and data wrangling, (2) quality checking and anomaly detection, (3) evidence extraction with sampling, (4) verification technologies (drone/CV), (5) prediction. Each stage builds the data discipline the next one depends on. Projects that jump straight to prediction on top of estimated progress automate their own delusion.

Chapter 15: EPC Best Practices — Lessons from Major Owner

Governance

Large owner organisations — national oil companies, major utilities, infrastructure authorities — run some of the most disciplined progress governance in the world, refined across hundreds of contractors. The practices below are generalised from that environment (no confidential material) and are worth adopting whatever your project's size.

The monthly reporting cycle as contract machinery

The mature pattern: a fixed cut-off date; contractor submission within a defined number of days (progress report, updated schedule in native format, S-curves, quantity backup); owner review meeting on a fixed calendar; formal comment and close-out. Two features make it work: the calendar never moves — the discipline of an immovable rhythm outweighs any single month's convenience — and submissions have defined content, so "the report" is a specification, not a surprise.

Contractor governance and verification rights

The Progress Measurement Procedure is a contract deliverable, approved before the first claim, with the measurement matrix and weight register as controlled appendices.

The owner holds explicit verification rights: access to registers, sampling walkdowns, audit of the roll-up arithmetic.

Progress for payment and progress for schedule control are reconciled — a divergence between claimed and control progress is itself a reportable item.

Schedule review as standing discipline

Owner schedule teams review contractor updates every period against quality criteria (logic integrity, constraint abuse, float patterns, out-of-sequence volume) before accepting the progress story the schedule tells. The generalisable lesson: a progress number is only as credible as the

schedule carrying it, and schedule quality checking is a monthly habit, not a claim-time forensic exercise.

Integrated progress: one system from engineering to commissioning

The best governance regimes require a single integrated progress framework across all phases and all contractors — common phase weights, common cut-offs, common roll-up rules — so programme-level progress means something. For contractors this arrives as burdensome standardisation; adopt it willingly. Fragmented measurement is what makes multi-contractor programmes ungovernable.

Stage gates and progress assurance

Major owners tie funding and phase authorisation to stage gates, each with progress-related entry criteria (engineering % before EPC award; MC % before commissioning mobilisation). Gates force the question "is the measured progress REAL?" at defined moments — with independent assurance reviews empowered to test the measurement system itself, not just the number.

Executive reporting upward

The reporting chain from project to corporate follows strict aggregation: every level sees the same underlying data, summarised — never re-authored. Where a project's numbers are re-negotiated at each management layer, the top of the house is managing fiction. The technical fix is the single-source dashboard architecture of Chapter 13; the cultural fix is leadership that punishes surprise more gently than it punishes concealment.

 **APEX TIP** Small contractors sometimes treat governance as big-company bureaucracy. The reverse is true commercially: a contractor who arrives with an owner-grade measurement system — procedure, matrix, registers, verification-ready records — is visibly cheaper to govern, and owners remember that at the next tender. Measurement discipline is a sales asset.

Chapter 16: Case Studies

Five condensed case studies, each built around one measurement lesson. Details are composites drawn from common industry situations; every pattern here recurs constantly across the project world.

Case 1 — EPC oil & gas plant: the front-loaded weightage

A gas processing EPC reported 38% at month 10 against a plan of 36% — comfortably green. A new project controls manager audited the weight register and found engineering weighted at 22% (against 12% by hours) — a legacy of tender-stage optimism never corrected. Restated, true progress was 31% against 36%. Painful month; but the restatement came early enough for recovery:

procurement acceleration and construction re-sequencing closed most of the gap by month 18. Lesson: audit the weight register at project start — inherited weights are inherited distortions, and the pain of restating grows every month it is deferred.

Case 2 — Refinery shutdown: measurement at shift speed

A 45-day turnaround ran progress at shift granularity: every work order carried planned hours, rules of credit ladders compressed to hours-scale milestones, and progress was captured at each shift handover with a hard 60-minute closing window. The measurement team's key design choice: NO estimated percentages allowed — every work order was 0/100 or milestone-laddered, because at shutdown speed there is no time to argue about "about 70%". Daily S-curves against the shift-level baseline caught a mechanical bottleneck on day 9 (exchanger bundle pulls running at 60% of planned rate); an extra night crew recovered it by day 14. Lesson: the faster the project, the simpler and harder-edged the rules must be — precision comes from frequency, not from decimals.

Case 3 — High-rise building: the repetition dividend

A 40-storey tower measured structure by floor-cycle (each floor a weighted package with a fixed internal ladder: formwork/rebar/pour/strike) and fit-out by room-completion counts per trade. Because every floor was near-identical, run-rate analytics became the control tool: planned cycle 6 days/floor, actuals plotted weekly. When cycle time crept from 6 to 8 days over floors 12–15, the trend was visible within three weeks and traced to crane congestion — resolved by adding a hoist and re-sequencing façade deliveries. Lesson: on repetitive work, measure the RATE as intensely as the total; the rate is the forecast.

Case 4 — Railway: interface progress across contractors

A metro line ran eight civil, track and systems contractors under one programme. Each contractor's internal progress was healthy; the programme still slipped — because handover interfaces (civil→track, track→systems access) carried no weight in anyone's system. The fix: an interface milestone register, each handover weighted in BOTH parties' systems and tracked at programme level, with access-readiness walkdowns as evidence. Slippage became visible at the interface where it originated, months earlier than before. Lesson: on multi-contractor programmes, measure the seams, not just the parts — programmes fail at handovers that no one's progress system owns.

Case 5 — Data centre: procurement-driven progress

A hyperscale data centre's fate rested on long-lead electrical equipment: generators, switchgear, UPS. The team weighted procurement at 45% of the project — unusual, and correct, because that is where the work content and risk sat. Package ladders emphasised the manufacturing span (Chapter 10) with monthly vendor-reported milestones verified by third-party inspection. When transformer manufacturing slipped six weeks at month 7, project-level progress showed it immediately — triggering an air-freight/re-sequencing decision while the site could still absorb it. A conventional construction-heavy weighting would have hidden the slip until it hit the site. Lesson: weight follows

work content wherever the content sits — in modern asset classes, that is increasingly the supply chain, not the field.

Chapter 17: The Template Library

Theory becomes practice through instruments. This chapter catalogues the working templates that accompany this guide — several ship with the Apex template pack today; the remainder are available through the Apex catalogue as they are released through e-mail.

The ten instruments

1. Progress Measurement Procedure (document template) — the controlled document that locks the system: methods, matrix, weights, cut-offs, verification, reporting. Structure follows this book's chapters.
2. Progress Rules Matrix — the Chapter 12 master table as a maintained register: work type, unit, basis, rule, evidence, source.
3. Weighting Register — every element's weight, basis and revision history; the auditable spine of the roll-up (Chapter 4).
4. Earned Value Worksheet — PV/EV/AC through SPI, CPI, EAC (three methods), ETC, VAC, TCPI in both RM and manhours; the Chapter 6 example, live. (Ships as Apex Template 04.)
5. Weekly Progress Report — the one-page discipline: executive summary, progress by discipline with auto roll-up, manpower, milestones. (Ships as Apex Template 03.)
6. Executive Dashboard — KPI cards, milestone trend and variance layout matching Chapter 13's design rules — Excel edition, with the Power BI build following the same specification.
7. Physical Progress Sheet — activity-level physical % computation from register roll-ups, formatted for direct P6 import (Chapter 5 workflow).
8. Quantity Tracking Sheet — controlled take-off totals, weekly installed capture with evidence references, run-rate analytics per commodity. (Chapter 11's backbone.)
9. S-Curve Workbook — plan/earned/forecast curves with variance and SPI, monthly locked-record discipline. (Ships as Apex Template 02; the companion Apex S-Curve guide documents the full method.)
10. Delay Register — contemporaneous delay event log: event, cause, affected activities, notices, progress impact — the record that connects measurement to entitlement.

Using the library

Adopt in this order on a new project: procedure shell (1) → matrix and weighting register (2, 3) → quantity tracker and physical progress sheet (8, 7) → reporting set (5, 9, 4) → dashboard (6) → delay register (10) from day one regardless. Adapt column structures to the contract, but preserve the disciplines the templates embody: controlled totals, locked history, evidence references, single source of truth.



Scan to access the obtain more templates — apexbusiness.training@gmail.com

A closing word

Progress measurement is unglamorous. No one celebrates the quantity tracker the way they celebrate a topping-out. But across every project in this book — real and composite — the difference between control and chaos was never the software, and rarely the intelligence of the team. It was whether the project could answer one question with evidence: how much is actually done? Build the system that answers it, run the cycle every period without exception, and you will spend your career being the person whose numbers end arguments rather than start them.

— Apex Business Solutions · Modern Project Controls Series